

Table 1. HarmBench refusal-type analysis using the three-class FalseReject rubric. USR is toxic-prompt Useful Safety Rate: Direct Refusal plus Safe Partial Compliance. Higher Safe Partial Compliance denotes more constructive safe engagement, although it can trade off against Direct Refusal; USR is the primary joint summary. Bold marks a best H_∞ result and underline marks the best competitor. Direct and Adversarial regimes receive equal weight, so the five jailbreak templates do not outweigh Direct prompts. Per method, the 1B/3B rows contain 240 Direct and 1,200 Adversarial outputs; the compact 8B row contains 120 Direct and 600 Adversarial outputs. Values are full-sample percentages $\pm$ ten-group behavior-clustered jackknife standard errors.

Model	Method	Direct refusal (%) $\uparrow$	Safe partial compliance (%) $\uparrow$	Full compliance (%) $\downarrow$	USR (%) $\uparrow$
Llama-3.2-1B-Instruct	Original	<u>91.83 $\pm$ 1.03</u>	0.92 $\pm$ 0.32	<u>7.25 $\pm$ 1.21</u>	<u>92.75 $\pm$ 1.21</u>
	A-LQR	90.00 $\pm$ 1.07	<u>2.17 $\pm$ 0.83</u>	7.83 $\pm$ 1.11	92.17 $\pm$ 1.11
	H$_\infty$ (ours)	96.25 $\pm$ 0.78	0.88 $\pm$ 0.46	2.88 $\pm$ 0.68	97.12 $\pm$ 0.68
Llama-3.2-3B-Instruct	Original	<u>82.33 $\pm$ 1.54</u>	3.54 $\pm$ 1.03	14.12 $\pm$ 1.18	85.88 $\pm$ 1.18
	A-LQR	81.62 $\pm$ 1.79	<u>4.58 $\pm$ 1.28</u>	<u>13.79 $\pm$ 1.32</u>	86.21 $\pm$ 1.32
	H$_\infty$ (ours)	91.79 $\pm$ 0.69	1.83 $\pm$ 0.63	6.38 $\pm$ 0.49	93.62 $\pm$ 0.49
Llama-3.1-8B-Instruct	Original	77.25 $\pm$ 3.40	5.58 $\pm$ 2.03	17.17 $\pm$ 2.61	82.83 $\pm$ 2.61
	A-LQR	<u>80.17 $\pm$ 3.35</u>	<u>6.33 $\pm$ 1.46</u>	<u>13.50 $\pm$ 3.09</u>	<u>86.50 $\pm$ 3.09</u>
	H$_\infty$ (ours)	91.33 $\pm$ 1.81	2.33 $\pm$ 0.64	6.33 $\pm$ 1.36	93.67 $\pm$ 1.36